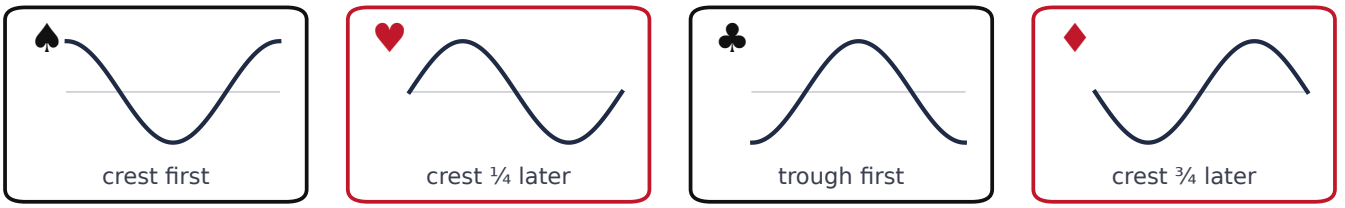
CATCH THE PHOTON · Wave Board

Each square shows the real wave you get by adding the cards played on a detector.



Same color, opposite suit (♠ and ♣, or ♥ and ♦): the waves cancel. Black with red: a quarter-step apart.

♠ one square up

32	25	20	17	16	17	20	25	32
25	18	13	10	9	10	13	18	25
20	13	8	5	4	5	8	13	20
17	10	5	2	1	2	5	10	17
16	9	4	1	0 START	1	4	9	16
17	10	5	2	1	2	5	10	17
20	13	8	5	4	5	8	13	20
25	18	13	10	9	10	13	18	25
32	25	20	17	16	17	20	25	32

♦ left

♥ right

♣ one square down

Number in a square = brightness = (height of the wave)². Blue token: Player A's detector. Purple token: Player B's detector.
A card that would push a token off the board cannot be played on that detector.

CATCH THE PHOTON · Rules

2 players · about 10 minutes · grades 6–8

You need

A deck of cards with the A–10 cards only (40 cards), the Wave Board, one blue and one purple token, a cup, and about 40 small chips or paper slips in blue and purple.

Setup

Both tokens start on START (brightness 0: dark). Shuffle. Deal 3 cards to each player.

Each round (play 8 rounds)

1. Both players secretly choose one card and lay it face-down beside a detector: your own (to make it brighter) or your opponent's (to cancel it).
2. Reveal together. Move that detector's token one square toward the card's suit: ♠ up, ♣ down, ♥ right, ♦ left. Card numbers do not matter.
3. Draw back up to 3 cards.

Catch the photon

After 8 rounds, read the number under each token. Put that many chips of each player's color in the cup. Draw one chip without looking: the photon lands on that detector, and that player wins. If both numbers are 0, play one more round.

Why it works

Every card is a wave arriving at a detector. Waves in step add up (bright). Opposite waves cancel (dark). Waves a quarter-step apart add a little. The brightness is the wave's height squared, and it sets the chance that the photon lands there. This is how real light and quantum computers work.

Try this

Put 3 ♥ and 4 ♠ on one detector. Where is the token? Its brightness is 25, because $3^2 + 4^2 = 5^2$ (Pythagoras).

Ghost Coin uses two phases (+ and −). Catch the Photon uses four, so waves can also meet at right angles.